

Nate Faulkenberry

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Senior Software Engineer | Full-Stack Web Development | AI-Assisted Software Engineering

SUMMARY

Senior software engineer specializing in full-stack web application development across higher education, enterprise software, consulting, and small-team environments. Experienced across the full software lifecycle, from requirements discovery and architecture through implementation, testing, deployment, and production support. Strong background in legacy modernization and AI-assisted software engineering.

CORE TECHNICAL SKILLS

Languages & Frameworks: TypeScript, JavaScript, React, Next.js, Node.js, HTML5, CSS3, PHP, Java, SQL

Backend & Data: PostgreSQL, MySQL, Prisma, tRPC, Redis, AWS S3, REST APIs, ETL/Data Integration

Infrastructure: Docker, Git, GitHub Actions, CI/CD, AWS, Heroku, Boomi

AI-Assisted Engineering: Coding Agents, Multi-Agent Workflows, Specification-Driven Development, Automated Testing, Codebase Analysis, Repository Automation

PROFESSIONAL EXPERIENCE

YALE UNIVERSITY – SCHOOL OF MANAGEMENT | Software Engineer 3

May 2019 – Present | New Haven, CT

- Built or substantially rebuilt approximately **10 production applications** supporting faculty, staff, students, and academic operations.
- Own projects across the full lifecycle, translating ambiguous requirements into architecture, implementation, testing, deployment, and production support.
- Established reusable engineering patterns around React, Next.js, TypeScript, Node.js, Docker, databases, APIs, and CI/CD.

- Currently integrate AI-assisted engineering into approximately **70% of development work**.

Course Auction: Led a year-long rewrite of a legacy .NET / SQL Server platform supporting hundreds of students, hundreds of courses, and approximately **5,000 bids per auction**. Rebuilt with Next.js, React, TypeScript, PostgreSQL, Prisma, and tRPC; reverse-engineered undocumented business rules, coordinated months of mock-auction testing, and oversaw the first production launch with no significant launch issues.

Yale Directory: Designed and built a replacement directory platform using Next.js, Node.js, Redis, PostgreSQL, Docker, and AWS S3. Implemented caching, asynchronous processing, fuzzy search, image processing, TensorFlow face detection, and a natural-language search prototype using embeddings, vector search, LangChain, and LLM integration.

Event Registration: Expanded an event-registration application into a generalized dynamic-form platform supporting approximately **100 forms/events and thousands of registrations**; integrated Salesforce Education Cloud and later migrated the frontend to Next.js/Docker.

Faculty Website Migration: Led migration of approximately **60 faculty websites**, including content migration, remediation, documentation, and training for support staff.

Additional: Rebuilt classroom digital signage with Next.js/React; developed Course Roster, Zoom Background Generator, TA Registration, EMBA Registration, Canvas Syllabus Link Sync, and a COVID-19 research application.

CAGE JSA | Web Consultant

2017 – 2019

Built WordPress websites, landing pages, marketing funnels, and consulting solutions while translating business requirements into production web experiences.

ENVISION PHARMA GROUP | Software Developer I

2014 – 2017

Developed and maintained enterprise pharmaceutical software across frontend and backend systems.

BENEFITSXML / SS&C TECHNOLOGIES | Front-End Web Developer

2010 – 2014

Developed interfaces and reusable frontend components for enterprise benefits administration applications serving **thousands of customers**.

ULTIMATE NUTRITION | Web Developer

2010

Developed and maintained web and e-commerce functionality.

EPATH LEARNING | Pro Services Web / Flash Designer & Developer

2008 – 2010

Delivered custom web and interactive learning experiences for clients.

AI-ASSISTED ENGINEERING

Developed a structured methodology for integrating AI coding agents into software engineering workflows beginning in 2025.

- Use Claude Code, OpenAI-based tools, and local models for implementation, testing, debugging, refactoring, codebase exploration, and documentation.
- Orchestrate multiple agents in parallel on larger development tasks, decomposing objectives into implementation, research, testing, and review.
- Establish project instructions, architecture documentation, implementation specifications, ADRs, and test plans for long-running agent workflows.
- Use agents for repository-wide refactoring, legacy modernization, large-codebase analysis, and automated test generation, including **hundreds of unit and regression tests**.
- Build supporting automation through GitHub Actions, repository tooling, development scripts, and reusable development workflows.

SELECTED INDEPENDENT ENGINEERING

AV Gen — Native C++ audiovisual application: 181,000+ lines of code and 2,200+ automated tests spanning audio analysis, procedural environments, animation, GPU rendering, and interactive 3D.

PX3 Synth — Native C++ audio software: Publicly released synthesizer/audio plugin for Logic and standalone use.

Additional work includes GitHub Actions automation, repository tooling, React Native, Flutter, Electron, browser applications, and music-theory software.

EDUCATION

New England Institute of Technology — Associate Degree, Multimedia & Web Design | 2007